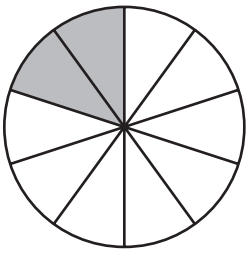
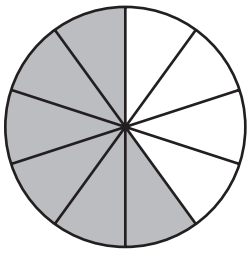
**1** In which fraction model does the shaded part represent the sum of this expression?

$$\frac{1}{10} + \frac{7}{10}$$

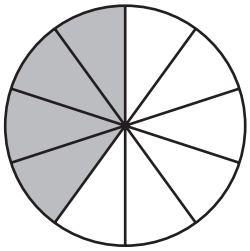
Ⓐ



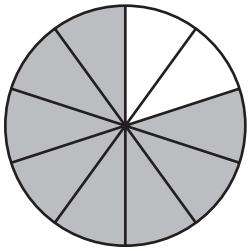
Ⓑ



Ⓒ



Ⓓ



**2** Takara is 4 years old. Her brother is 2 times as old as Takara.

Which equation can be used to find $b$, the age in years of Takara's brother?

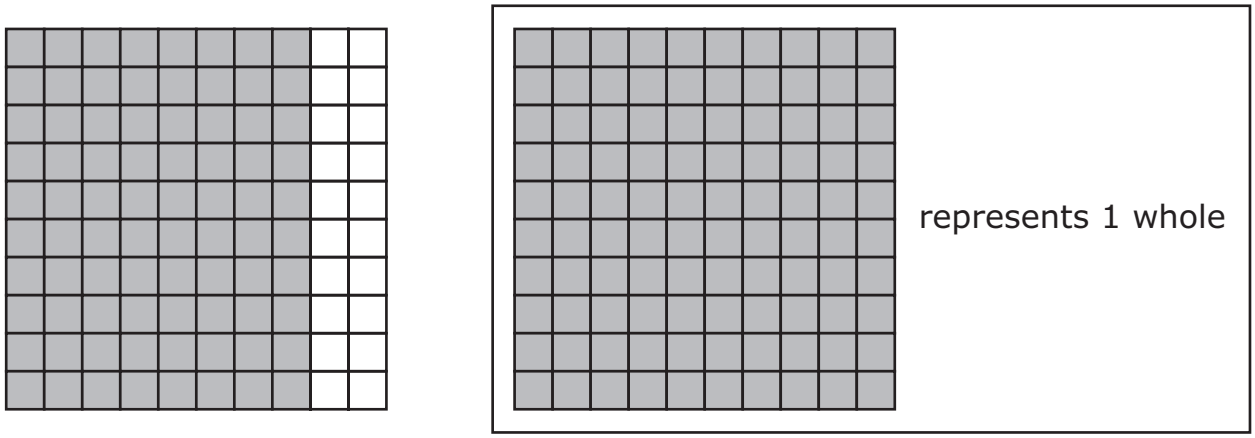
Ⓐ $b = 1 \times 2$

Ⓑ $b = 2 \times 2$

Ⓒ $b = 4 \times 2$

Ⓓ $b = 8 \times 2$

**3** The shaded portion of this model represents a fraction less than 1.




Which of these decimals are equivalent to the fraction represented in the model?

Select the **two** correct answers.

Ⓐ 0.80

Ⓑ 8.0

Ⓒ 0.08

Ⓓ 80.0

Ⓔ 0.8

**4** Ms. Lewis bought 5 sticks of butter. Each stick of butter weighs $\frac{1}{4}$ pound.

What is the total weight, in pounds, of the 5 sticks of butter?

Ⓐ $\frac{5}{4}$

Ⓑ $\frac{6}{4}$

Ⓒ $\frac{1}{20}$

Ⓓ $\frac{5}{20}$

**5** Which of these numbers are prime numbers?

Select the **two** correct answers.

Ⓐ 27

Ⓑ 43

Ⓒ 52

Ⓓ 67

Ⓔ 95

**This question has four parts. Be sure to label each part of your response.**

**6** A. Write this number in standard form.

| one hundred sixty-eight thousand, twenty-one |
|---|

B. Write the number 168,201 in expanded form.

C. Use >, <, or = to write a comparison of the two numbers in Part A and Part B. Explain how you got your answer.

D. Write a number that is 1,000 greater than the number 169,201. Explain how you got your answer.

**Write your answers on the next page.**

**7** A sign is in the shape of a square. Each side of the sign has a length of 6 inches.

What is the area of the sign?

Ⓐ 12 square inches

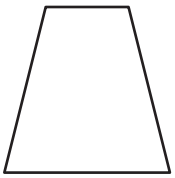
Ⓑ 24 square inches

Ⓒ 36 square inches

Ⓓ 48 square inches

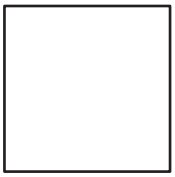
**8** Which of these shapes have **at least** one line of symmetry?

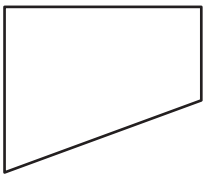
Select the **three** correct answers.

Ⓐ 

Ⓑ

Ⓒ

Ⓓ 

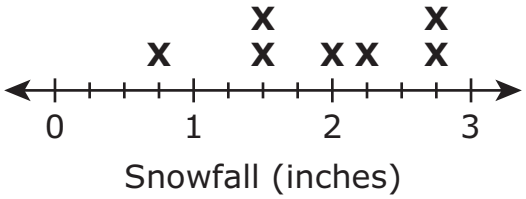
Ⓔ 

**9** This list shows the snowfall amounts, in inches, for eight towns during a storm.

$2\frac{1}{4}$, $1\frac{1}{2}$, $2$, $2\frac{3}{4}$, $1\frac{1}{2}$, $\frac{3}{4}$, $2\frac{3}{4}$, $1\frac{1}{4}$

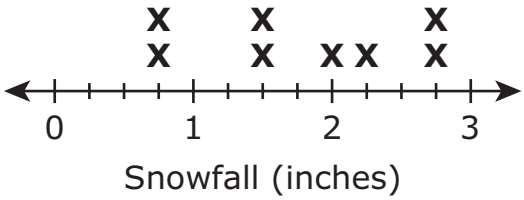
This line plot also shows some of the snowfall amounts. One of the snowfall amounts is missing from the line plot.

**Snowfall Amounts**

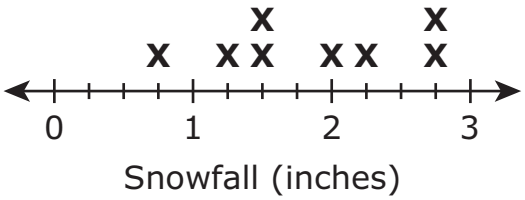



Which of these line plots shows all eight snowfall amounts correctly plotted?

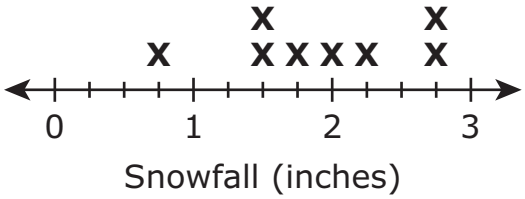
Ⓐ **Snowfall Amounts**



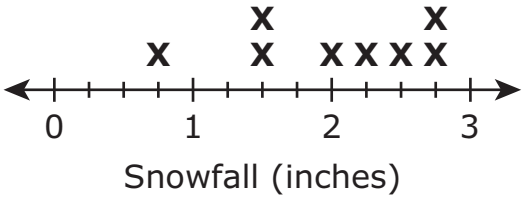

Ⓑ **Snowfall Amounts**




Ⓒ **Snowfall Amounts**




Ⓓ **Snowfall Amounts**



**10** What is the value of $\boxed{?}$ that makes this number sentence true?

$$325 = 147 + \boxed{?}$$

Ⓐ 178

Ⓑ 188

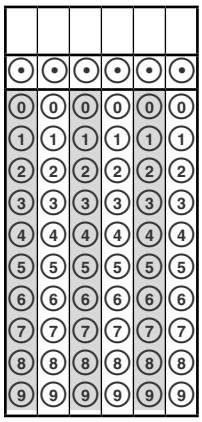
Ⓒ 222

Ⓓ 472

**11** The rule for a pattern is "multiply by 3." The first number in the pattern is 2.

What is the **fifth** number in the pattern?

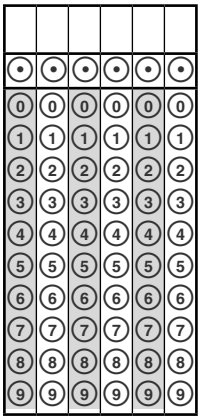
Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



**12** What value of $x$ makes this equation true?

$$\frac{7}{10} = \frac{x}{100}$$

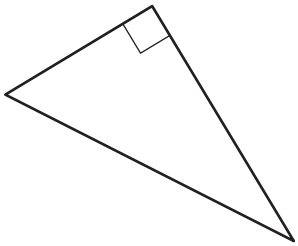
Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



**13** Which of these triangles have at least one obtuse angle?

Select the **two** correct answers.

Ⓐ



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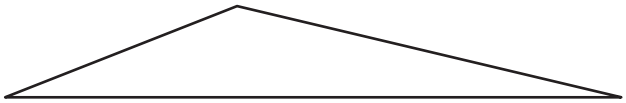

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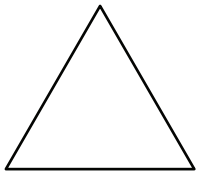

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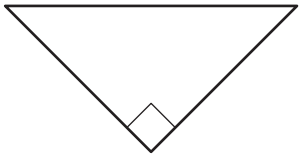
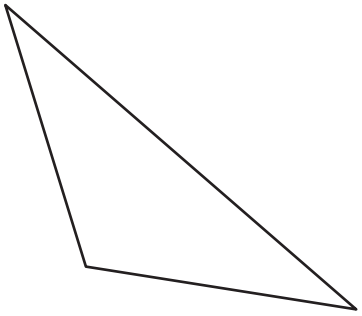


Ⓔ



**15** Alonzo and Mindy are buying pretzels to share with their class. They will put all the pretzels into bags.

- Alonzo buys 8 pretzels.
- Mindy buys 3 times as many pretzels as Alonzo.
- Each bag will hold up to 5 pretzels.

Alonzo and Mindy want to know the least number of bags they need to hold all the pretzels.

Which of these statements are true?

Select the **two** correct answers.

Ⓐ Alonzo and Mindy have a total of 24 pretzels.

Ⓑ Alonzo and Mindy have a total of 32 pretzels.

Ⓒ Alonzo and Mindy need 5 bags to hold all the pretzels.

Ⓓ Alonzo and Mindy need 6 bags to hold all the pretzels.

Ⓔ Alonzo and Mindy need 7 bags to hold all the pretzels.

**This question has two parts.**

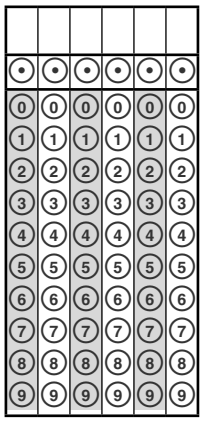
**16** A shopper bought peppers and grapes.

**Part A**

The shopper bought $\frac{54}{100}$ pound of peppers.

What is the decimal equivalent of the fraction $\frac{54}{100}$?

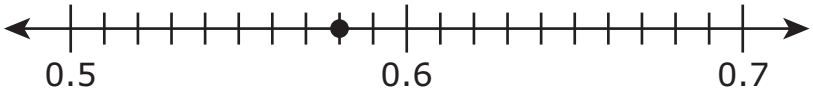
Enter your answer in the answer boxes at the top of the answer grid **and** completely fill the matching circles.



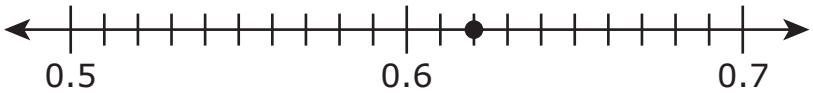
**Part B**

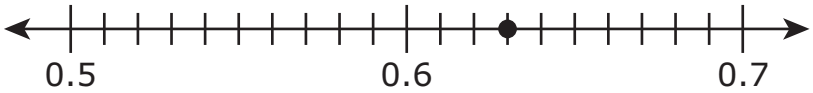
The shopper bought 0.62 pound of grapes.

Which of these number lines shows a point that represents where 0.62 is located?

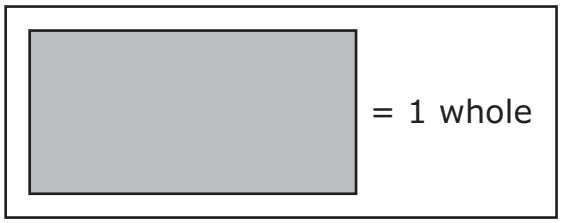
Ⓐ 


Ⓑ


Ⓒ 

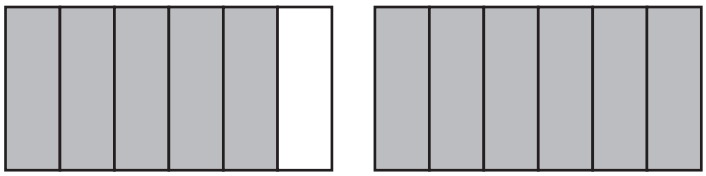

Ⓓ 

**17** A model of 1 whole is shown.

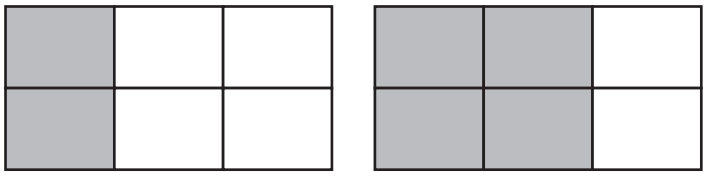



Based on the model, which of these models is shaded to represent a fraction that is equivalent to $1\frac{2}{3}$?

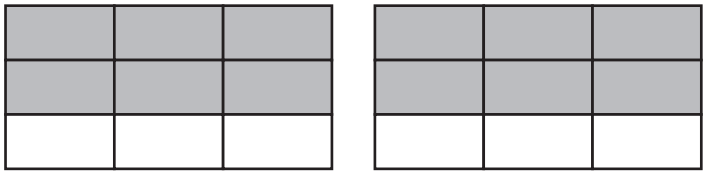
Ⓐ



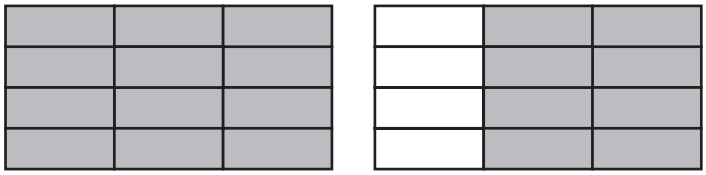
Ⓑ



Ⓒ



Ⓓ



**18** Which of these numbers round to 30,000 when rounding to the nearest **ten thousand**?

Select the **two** correct answers.

Ⓐ 34,124

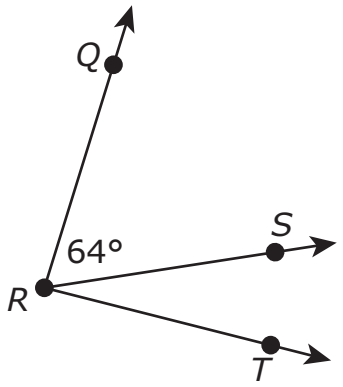
Ⓑ 35,021

Ⓒ 37,826

Ⓓ 32,788

Ⓔ 36,249

**19** Angle *QRS* has a measure of 64°, as shown in this diagram.




Angle *QRT* has a measure of 87°. What is the measure of angle *SRT*?

Ⓐ 23°

Ⓑ 26°

Ⓒ 64°

Ⓓ 151°

**20** A parent bought equal amounts of blue cloth and yellow cloth to make a costume. The parent used $\frac{3}{6}$ of the blue cloth and $\frac{4}{12}$ of the yellow cloth.

Which of these number sentences correctly compare the fractions $\frac{3}{6}$ and $\frac{4}{12}$?

Select the **two** correct answers.

Ⓐ $\frac{3}{6} > \frac{4}{12}$

Ⓑ $\frac{3}{6} < \frac{4}{12}$

Ⓒ $\frac{4}{12} = \frac{3}{6}$

Ⓓ $\frac{4}{12} > \frac{3}{6}$

Ⓔ $\frac{4}{12} < \frac{3}{6}$